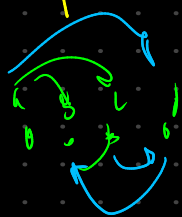
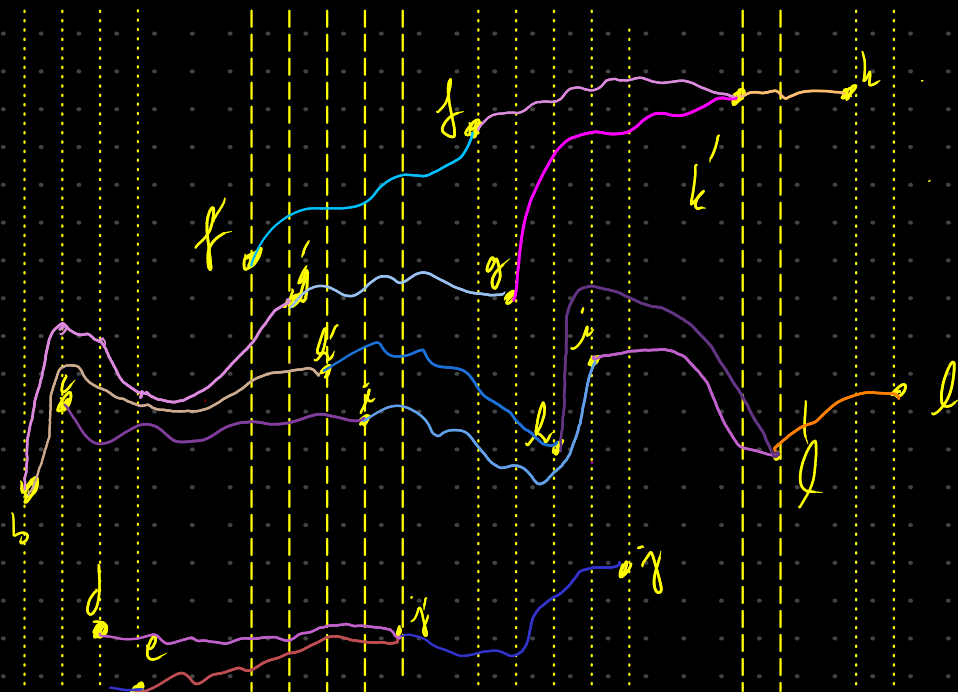


Let's add one edge to the counter example graph

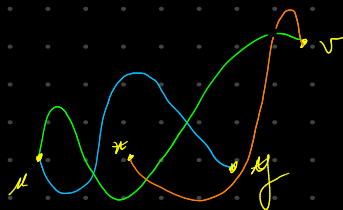
(a, d)



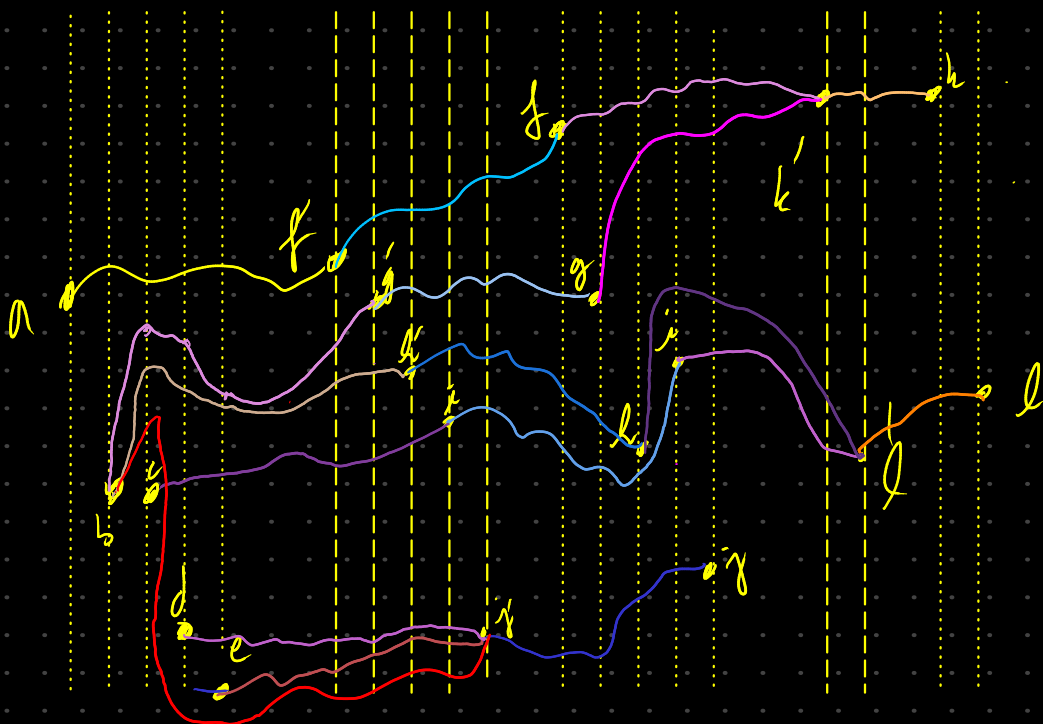
$$\begin{aligned} \varphi(a,b) &= 1 & \varphi(a,c) &= \times & \varphi(a,d) &= 1 & \varphi(a,e) &= 1 \\ \varphi(b,c) &= \times & \varphi(b,d) &= 0 & \varphi(b,e) &= 0 \\ \varphi(c,d) &= 1 & \varphi(c,e) &= 1 & \varphi(d,e) &= \times \\ \varphi(f,g) &= \times & \varphi(f,h) &= 1 & \varphi(f,i) &= \times & \varphi(f,j) &= 1 \\ \varphi(g,h) &= 1 & \varphi(g,i) &= 1 & \varphi(g,j) &= 0 & \varphi(h,i) &= 1 \\ \varphi(h,j) &= 0 & \varphi(i,j) &= 1 \end{aligned}$$

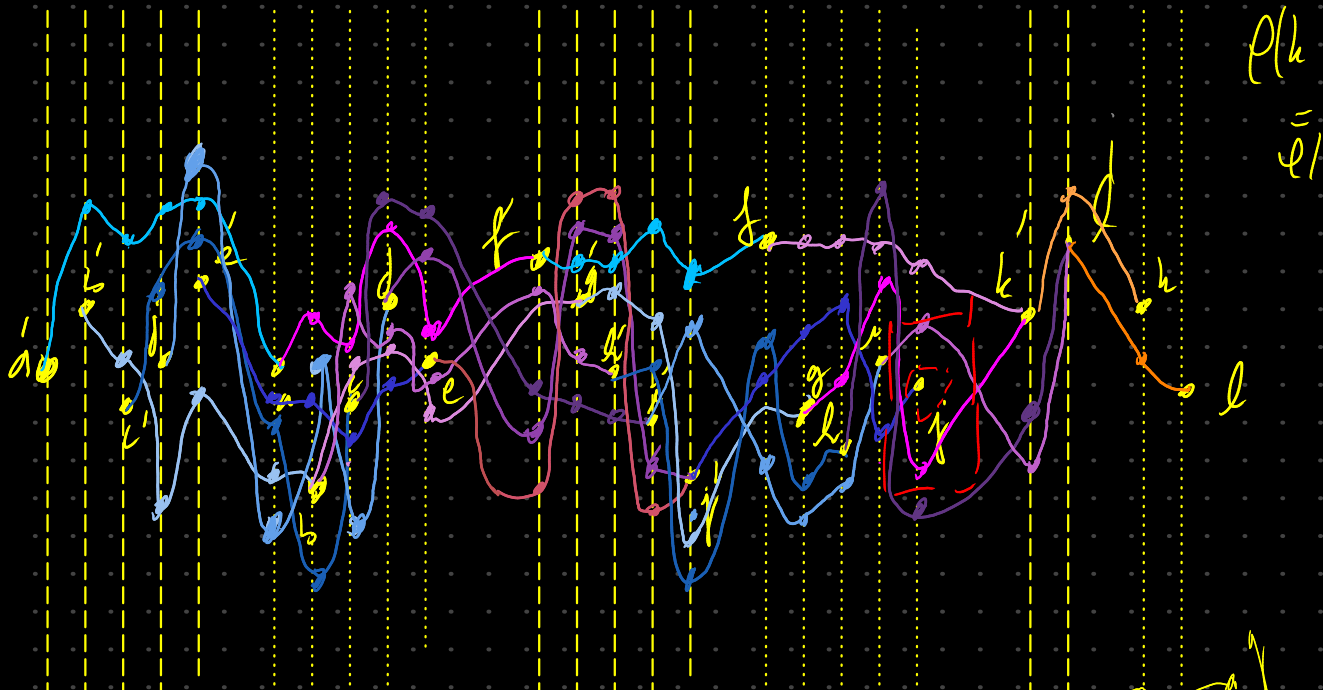


What if we actually have a cycle

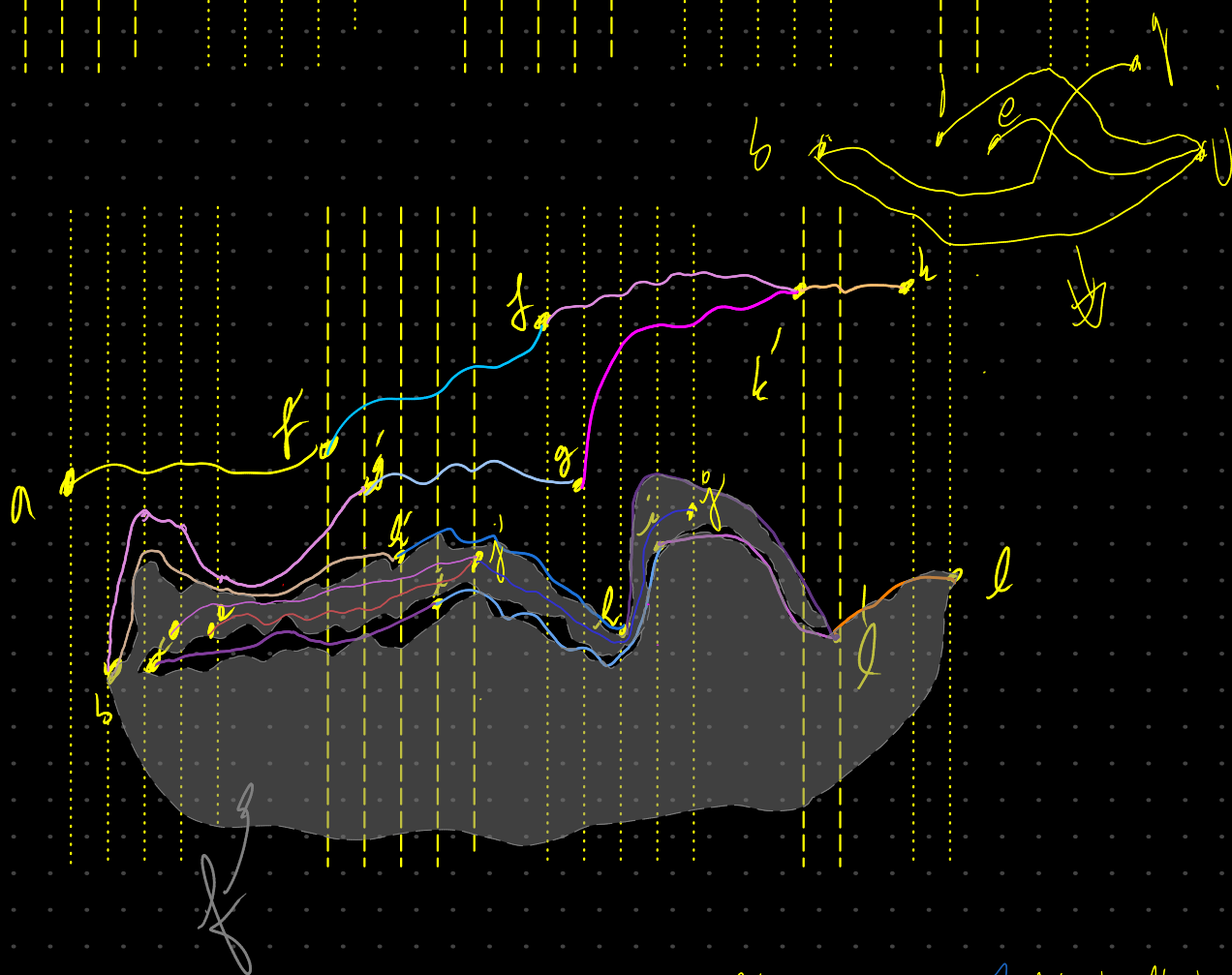


$$\begin{aligned} \varphi(c,d) &= 1 \\ \varphi(c,e) &= 1 \\ \varphi(d,e) &= 1 \\ \varphi(b,d) &= 0 \\ \varphi(b,e) &= 0 \\ \varphi(b,c) &= 1 \end{aligned}$$





$\rho(h)$
 $\bar{\rho}(l)$

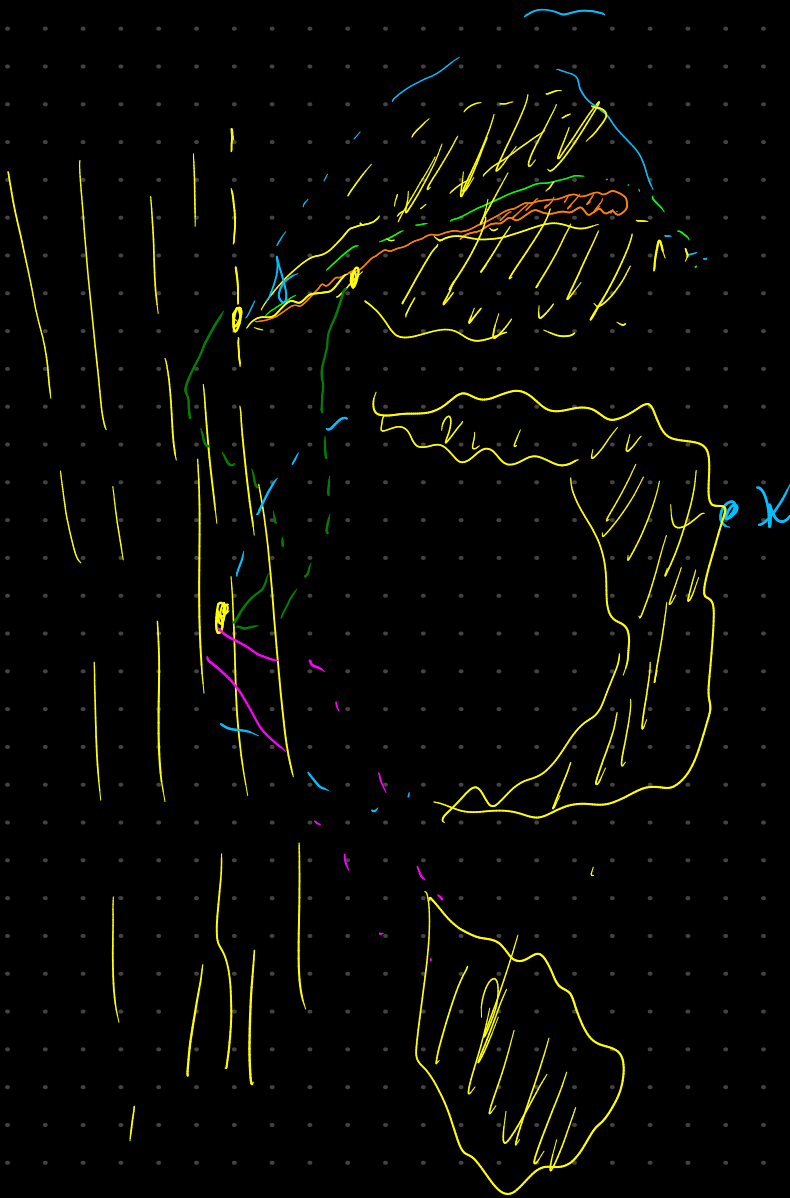
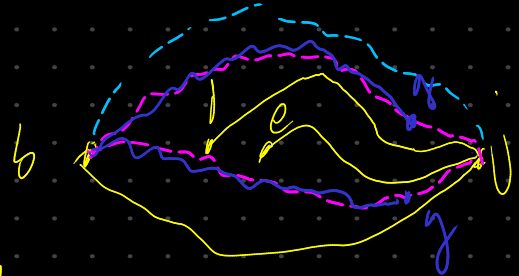
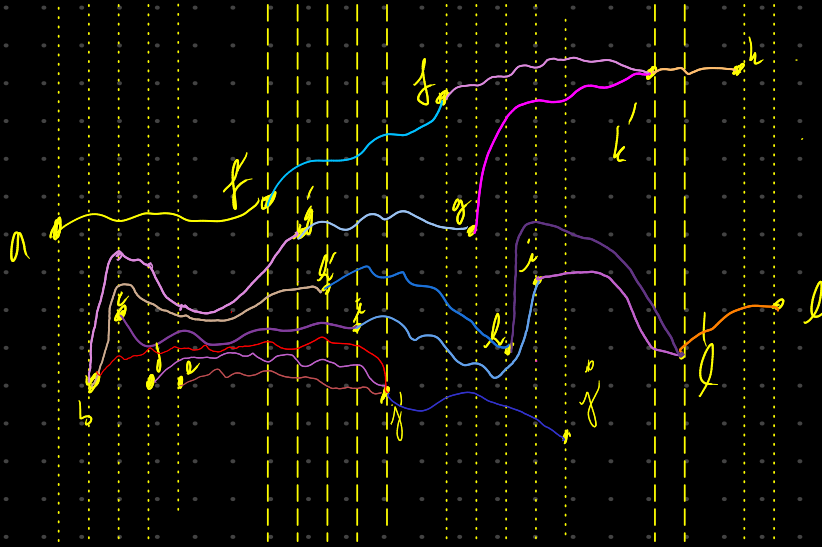
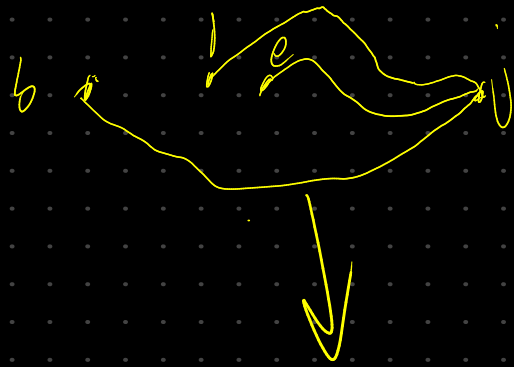


$$\begin{aligned} \varphi(ab) &= 1 \Rightarrow 0 & \varphi(ea) &= 0 \\ \varphi(ua) &= 0 & \varphi(eb) &= 1 \Rightarrow 0 \\ \varphi(ul) &= 0 & \varphi(ec) &= 0 \\ \varphi(ue) &= 1 & \varphi(ed) &= 0 \end{aligned}$$

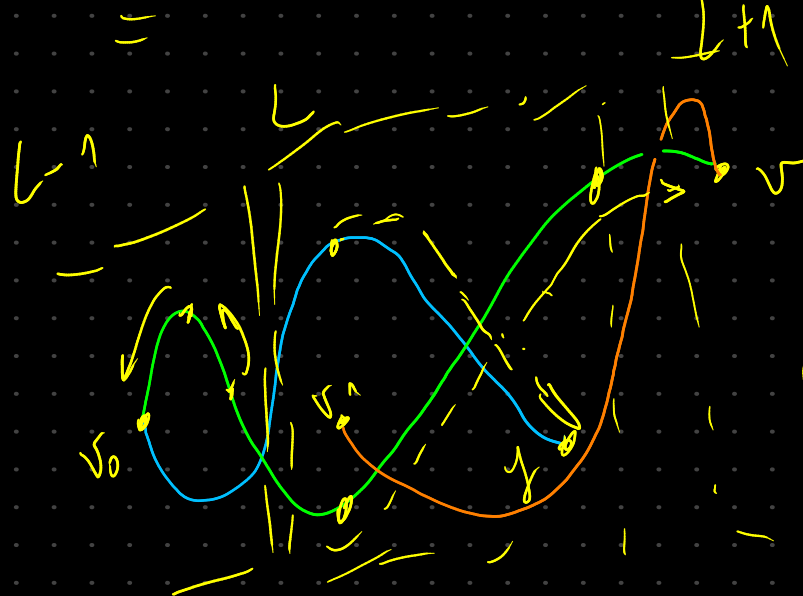
φ

$$\begin{aligned} \varphi(ab) &= 1 & \varphi(ac) &= \times & \varphi(ad) &= 1 & \varphi(ae) &= 1 \\ \varphi(bd) &= \times & \varphi(bd) &= 0 & \varphi(be) &= 0 \\ \varphi(cd) &= 1 & \varphi(ce) &= 1 & \varphi(de) &= \times \\ \varphi(fg) &= \times & \varphi(fh) &= 1 & \varphi(fi) &= \times & \varphi(fj) &= 1 \\ \varphi(ga) &= 1 & \varphi(gi) &= 1 & \varphi(gj) &= 0 & \varphi(hi) &= 1 \\ \varphi(hj) &= 0 & \varphi(is) &= 1 \end{aligned}$$

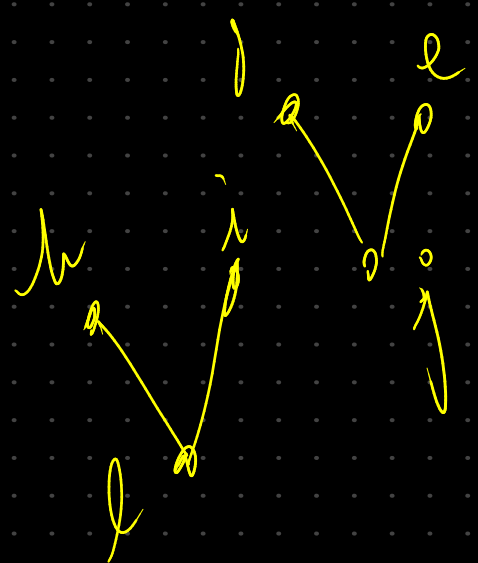
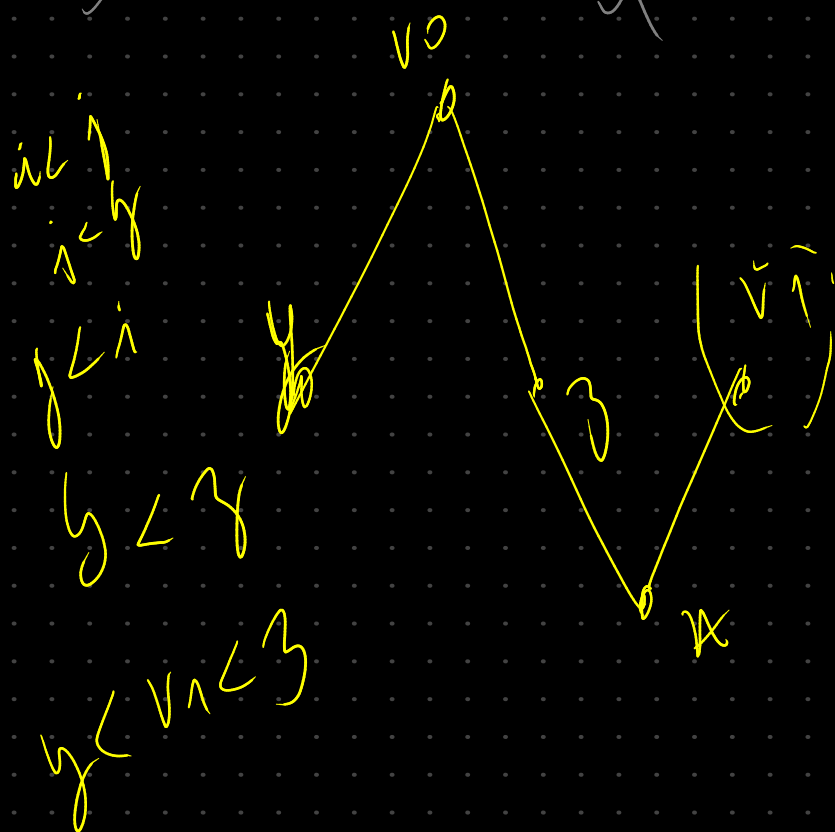
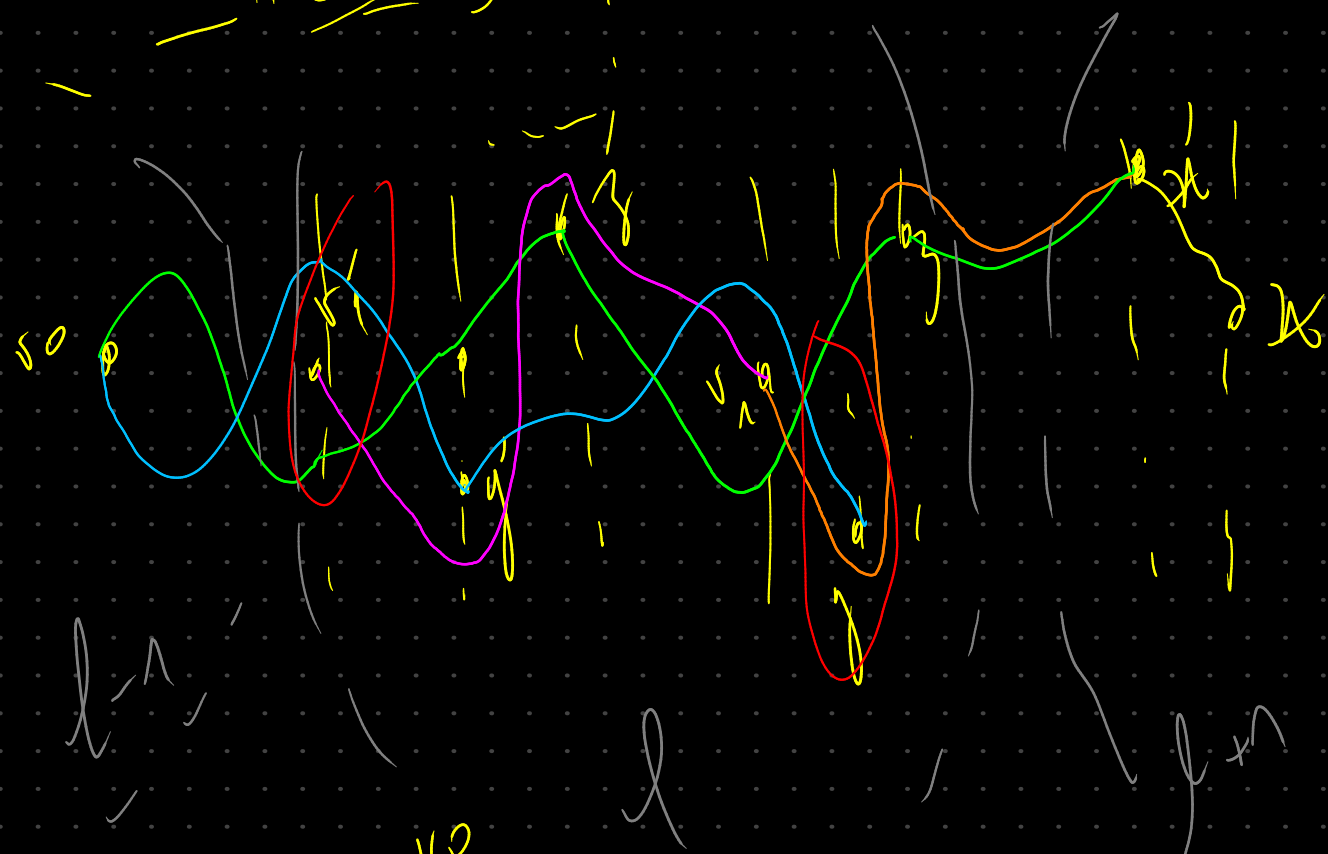
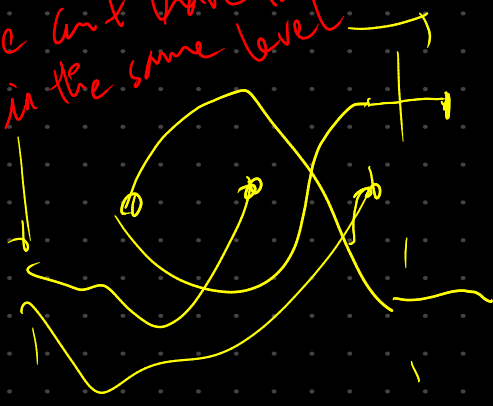
$$\begin{aligned} \varphi(\hat{x}\hat{u}) &= 0 & \varphi(\hat{u}\hat{h}) &= 1 \\ \varphi(\hat{u}\hat{w}) &= 0 & \varphi(\hat{u}\hat{w}) &= 1 \end{aligned}$$



Handwritten text in blue ink: $x > 5$ and $x \in \mathbb{N}$.



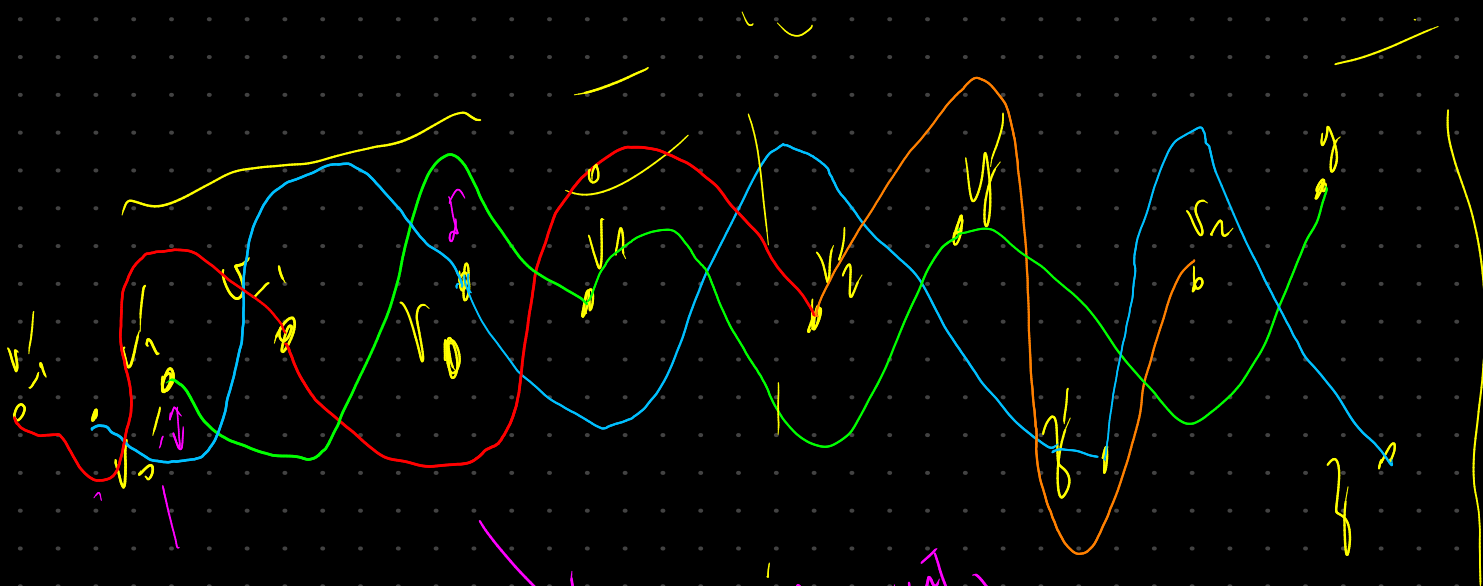
We will drive this in the same level.



I think I found the relation

$$v_1 \downarrow v_0 \downarrow y < z$$

$$v_2 \downarrow z < v_2 < y$$

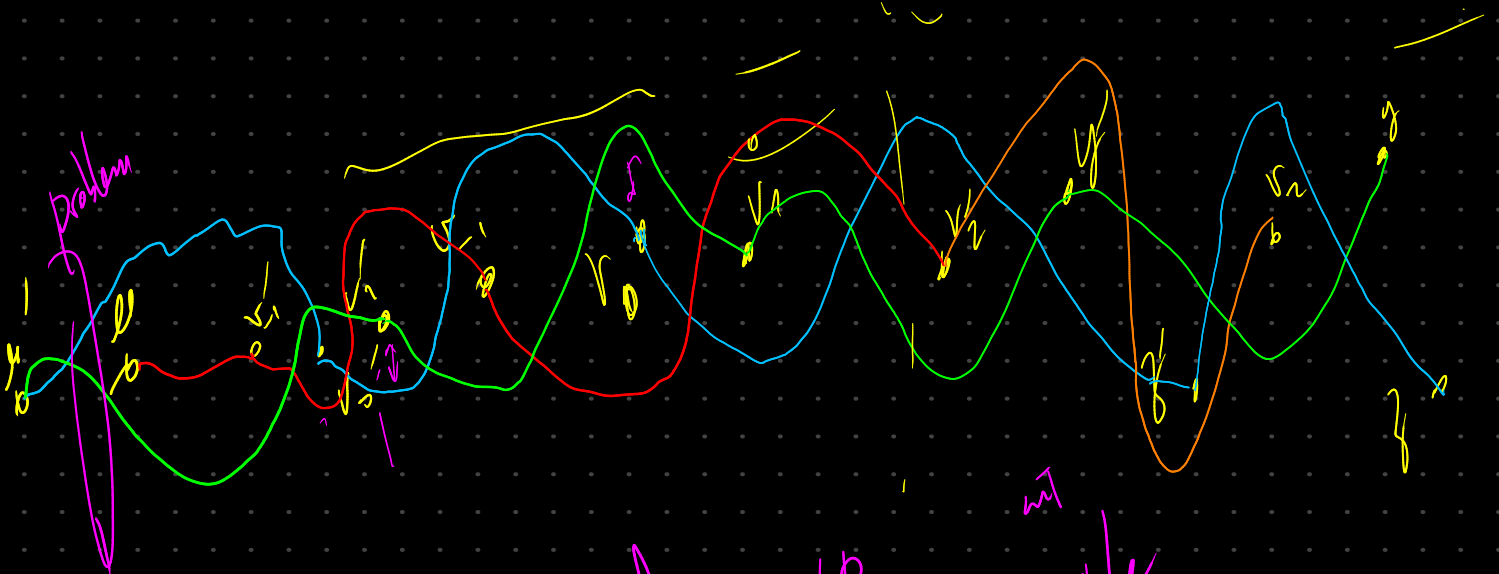


$$\psi(v_1, v_1) = 1$$

$$\psi(v_1, v_0) = 0$$

$$\psi(v_1, v_2) = 0$$

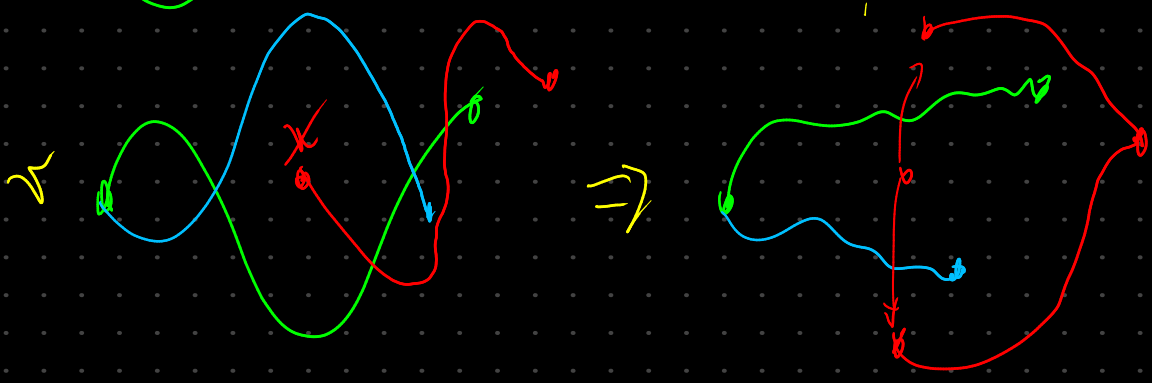
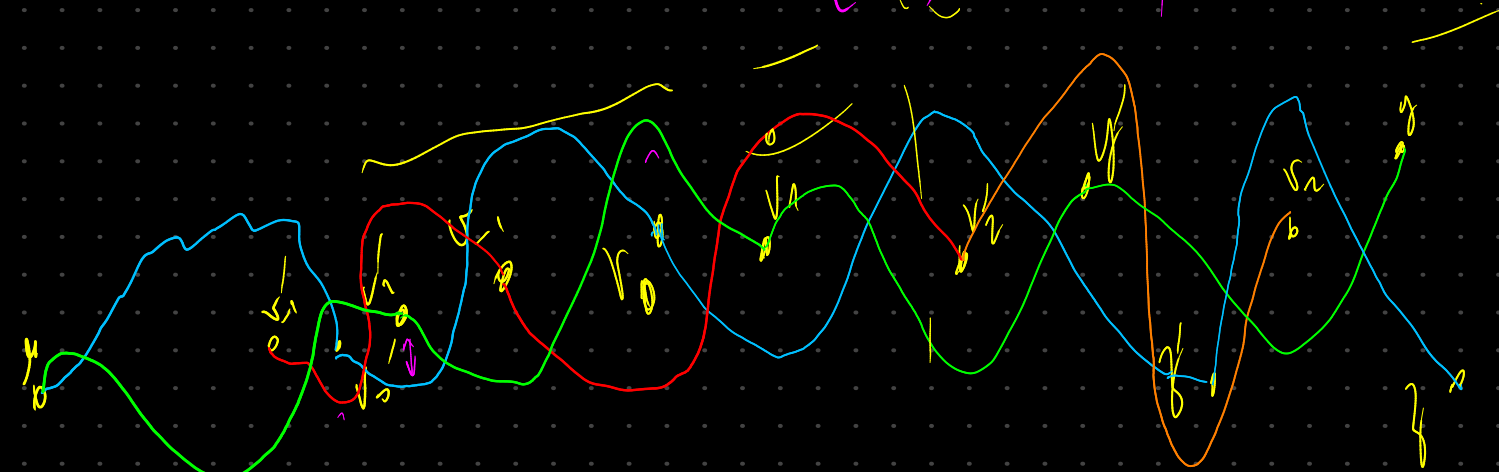
it's not relevant
if the correct
limits don't
form and don't
work at the end.
(we can draw them)
independently

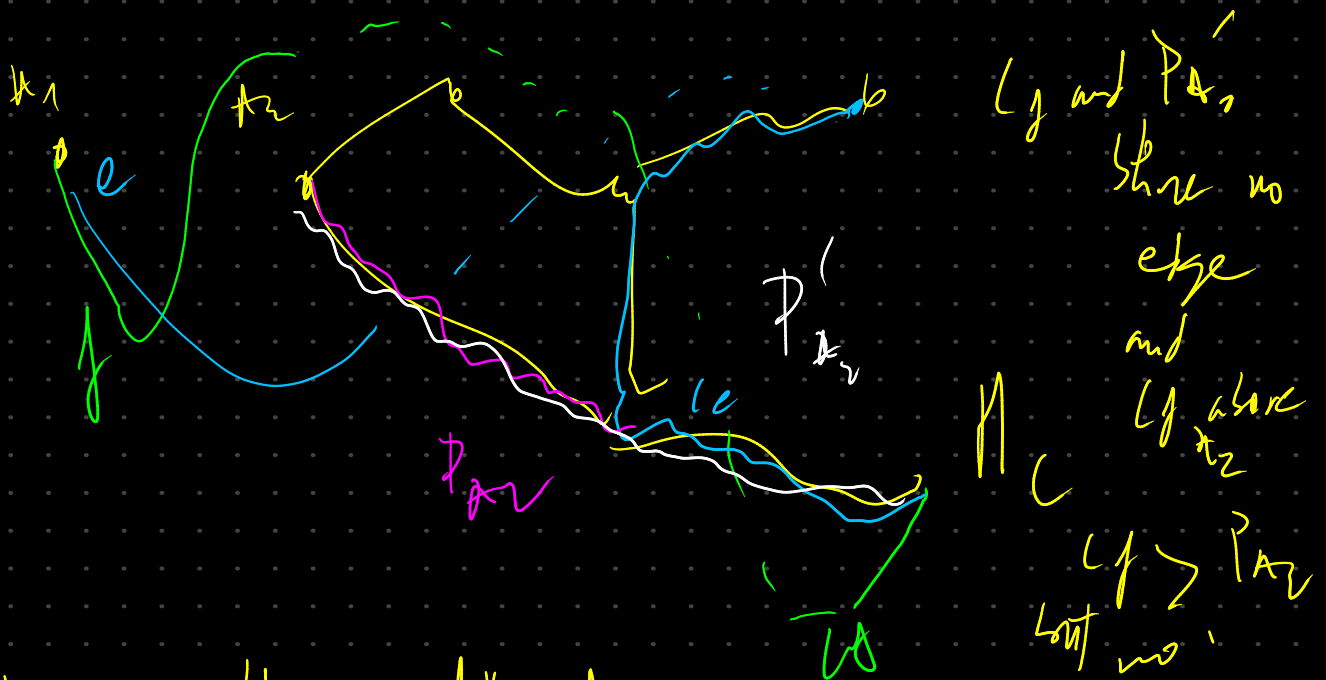


Problem

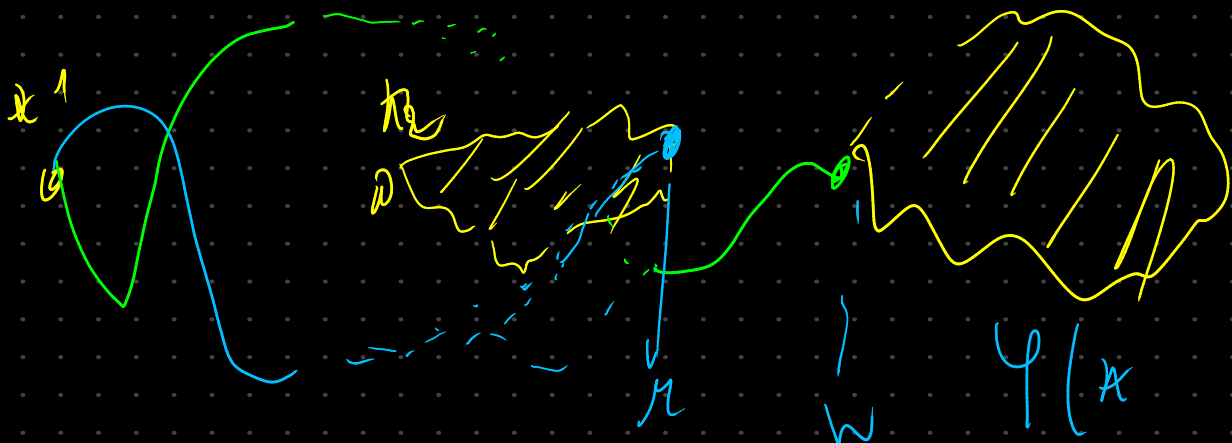
link from stretch edge

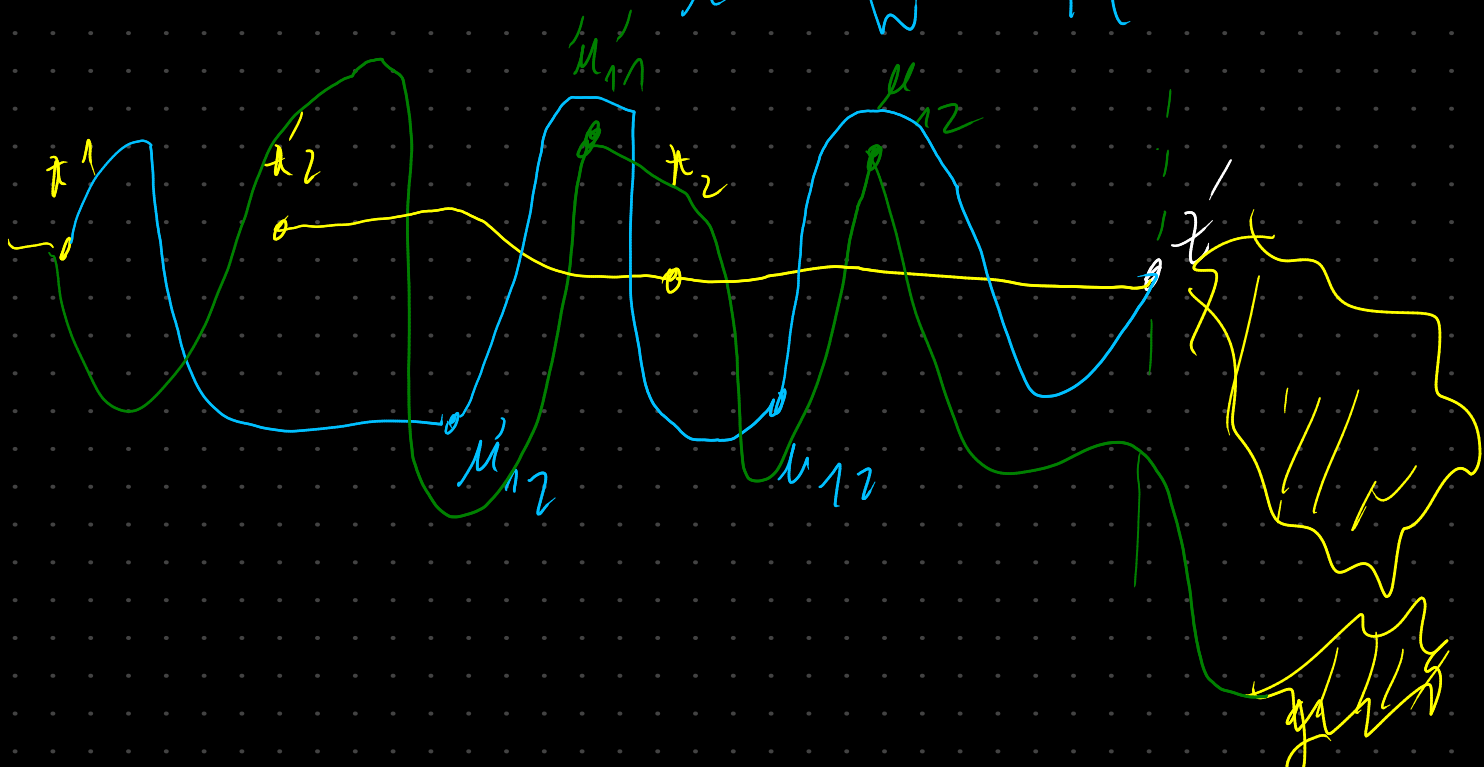
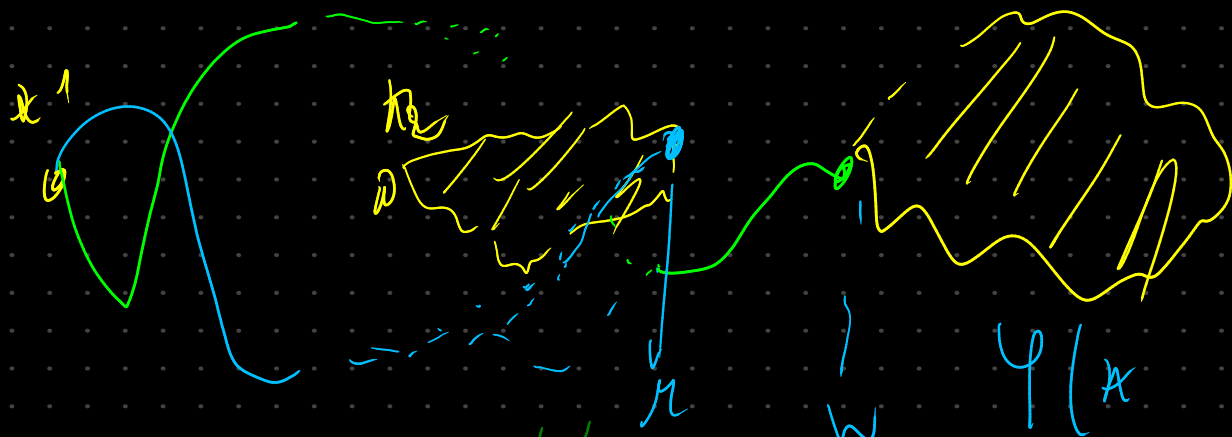
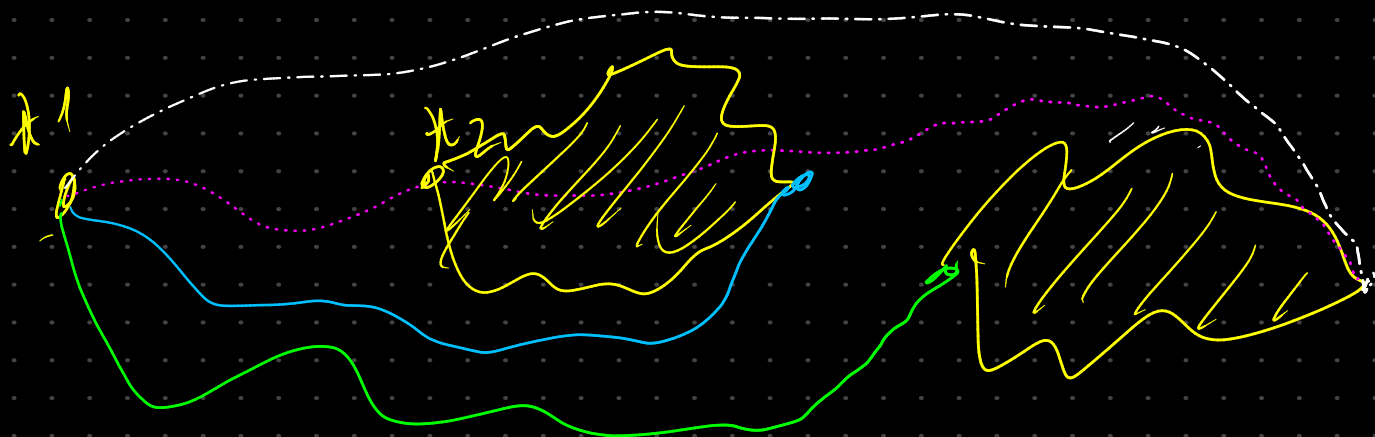
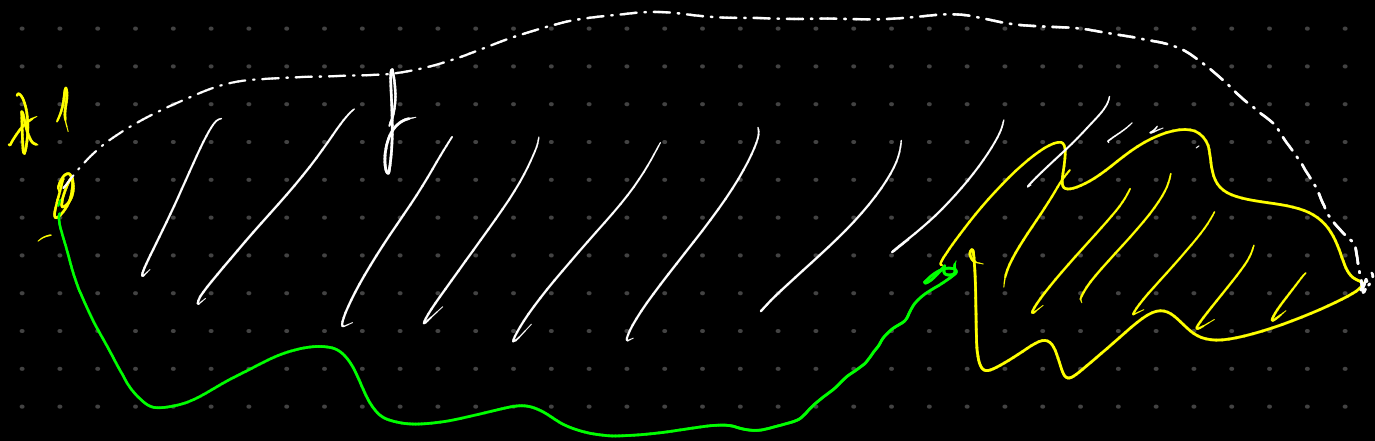
we can have the example
 leading to lemma 2.3 only
 edges linking $L(l(v)) \neq L(l(w))$
 $(v, w) \in E(w)$



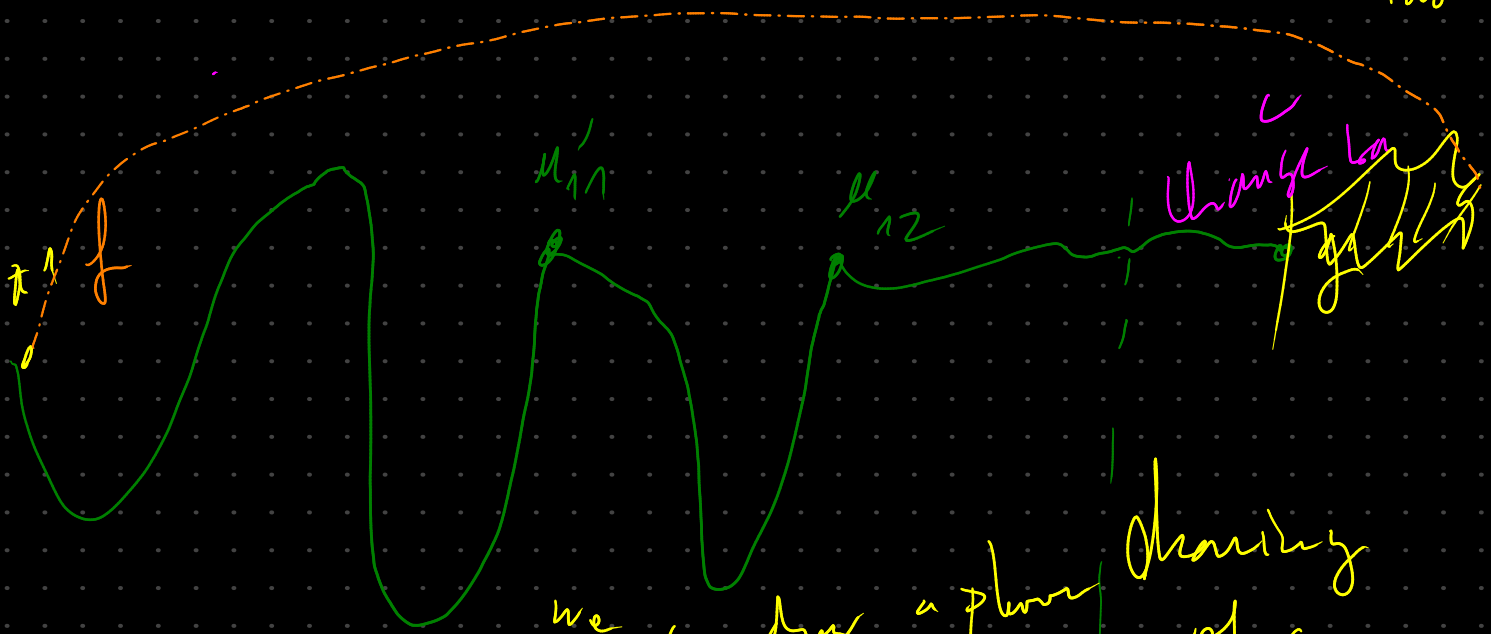


what would be diff for multiple lamps?

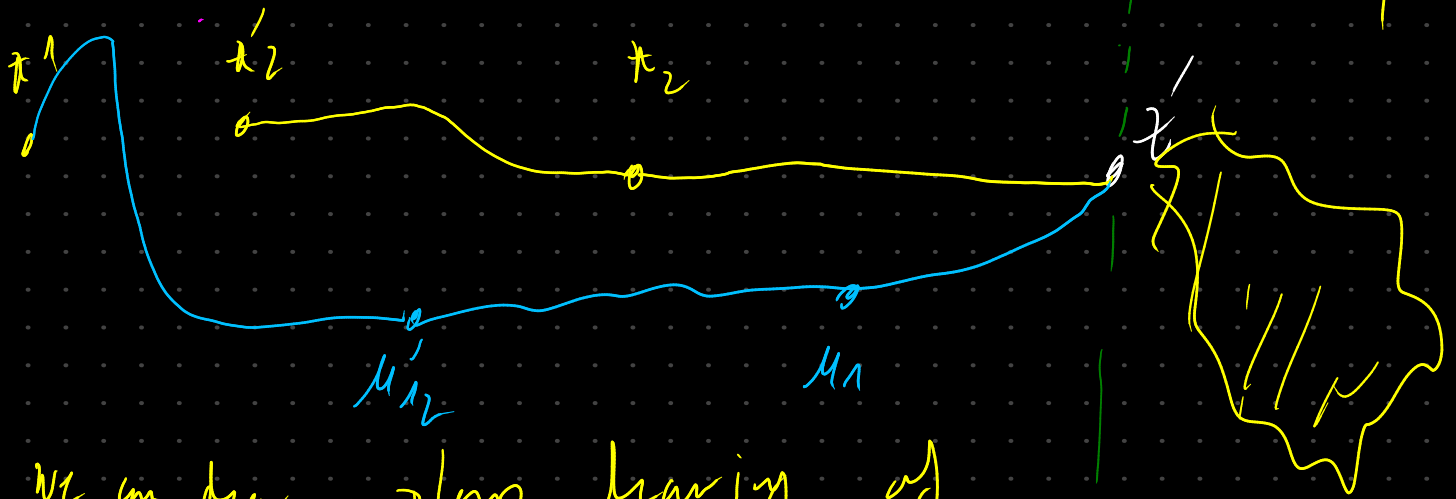




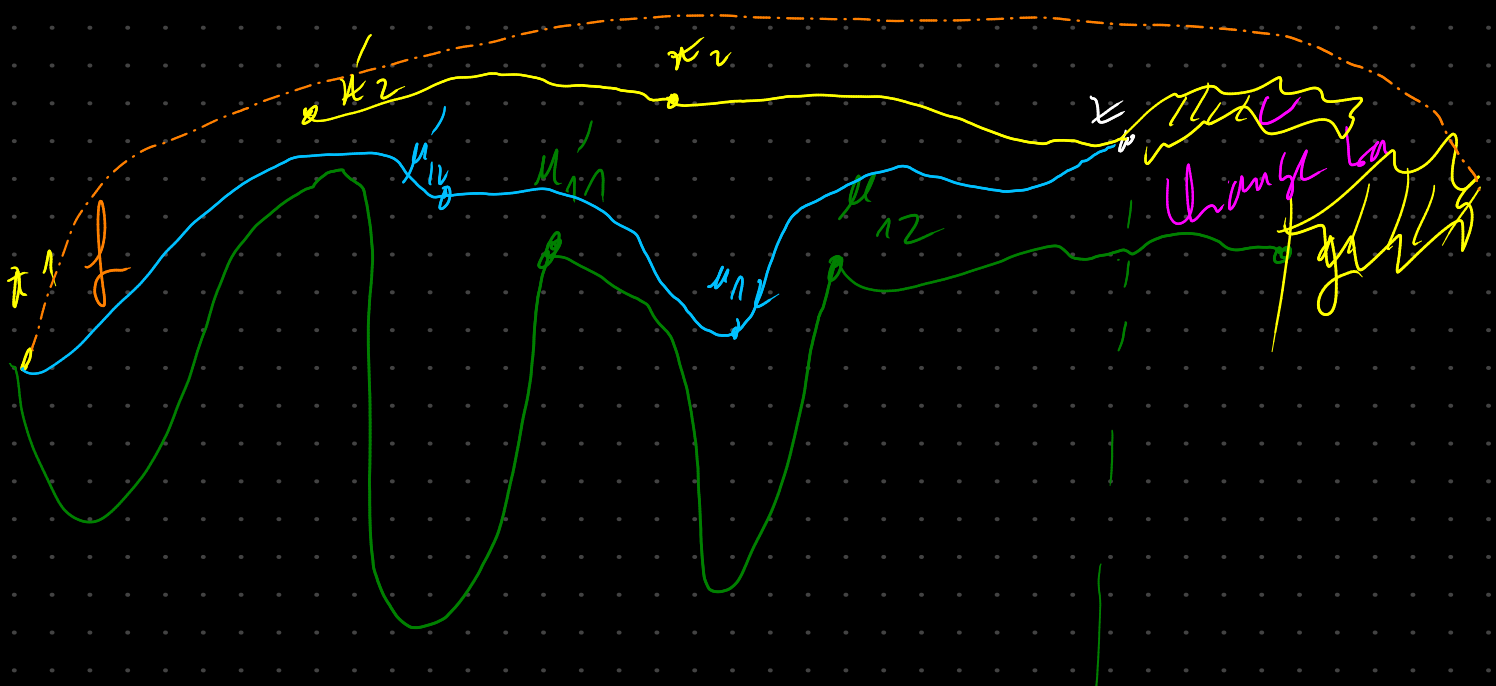
let's assume given H has π -max vertex b/w the two.



we can draw a planar drawing of green H



we can draw a planar drawing of blue H



entanglement can be removed only with edges $e(u, v)$ with $L(u) \neq L(v)$

